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Search

Table name	Total Size	Tuples inserted	Tuples updated	Tuples deleted	Tuples HOT updated	Live tuples
airline	24 KB	5	0	0	0	5
airport	24 KB	5	0	0	0	5
baggage	24 KB	5	0	0	0	5
baggage_check	24 KB	5	0	0	0	5
boarding_pass	24 KB	5	0	0	0	5
booking	24 KB	5	0	0	0	5
booking_flight	24 KB	5	0	0	0	5
flights	32 KB	5	0	0	0	5
passengers	24 KB	5	0	0	0	5
security_check	24 KB	5	0	0	0	5

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Object Editor: labwork/postgres@PostgreSQL 18

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 - Tables (5)
 - airline
 - airport

Query: `SELECT * FROM airline`

Data Output: Messages Notifications

Showing rows: 1 to 5 of 1

	airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	1	KZ	Air Astana	Kazakhstan	2026-09-23 00:04:24.805474	2026-09-23 00:04:24.805474
2	2	FS	FlyArystan	Kazakhstan	2026-09-23 00:04:24.805474	2026-09-23 00:04:24.805474
3	3	TK	Turkish Airlines	Turkey	2026-09-23 00:04:24.805474	2026-09-23 00:04:24.805474
4	4	EK	Emirates	UAE	2026-09-23 00:04:24.805474	2026-09-23 00:04:24.805474
5	5	LH	Lufthansa	Germany	2026-09-23 00:04:24.805474	2026-09-23 00:04:24.805474

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Query

```
SELECT * FROM flights
```

Data Output

	flight_id [PK] integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying
1	1	2026-09-25 08:00:00	2026-09-25 09:00:00	1	2	A1	B2
2	2	2026-09-25 10:00:00	2026-09-25 13:00:00	1	3	A2	C1
3	3	2026-09-26 12:00:00	2026-09-26 15:00:00	2	4	B1	D2
4	4	2026-09-26 16:00:00	2026-09-26 20:00:00	3	5	C2	E1
5	5	2026-09-27 09:00:00	2026-09-27 14:00:00	4	1	D1	A3

Total rows: 5 Query complete 00:00:00.044 CRLF Ln 1, Col 22

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Query

```
1 SELECT * FROM booking
```

Query History

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)
1	1	1	1	Website	2026-09-23 00:07:41.495996	2026-09-23 00:07:41.495996	Confirmed
2	2	2	2	Mobile App	2026-09-23 00:07:41.495996	2026-09-23 00:07:41.495996	Confirmed
3	3	3	3	Website	2026-09-23 00:07:41.495996	2026-09-23 00:07:41.495996	Pending
4	4	4	4	Mobile App	2026-09-23 00:07:41.495996	2026-09-23 00:07:41.495996	Confirmed
5	5	5	5	Website	2026-09-23 00:07:41.495996	2026-09-23 00:07:41.495996	Cancelled

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 - Columns
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 - sch_dep
 - sch_arr
 - departir
 - arriving
 - departir
 - arriving
 - airline_i
 - act_dep

Dashboard Properties SQL Statistics labwork/postgres@PostgreSQL 18*

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Query Query History

```
1 SELECT
2   tc.table_name,
3   tc.constraint_name,
4   kc.column_name
5 FROM information_schema.table_constraints tc
6 JOIN information_schema.key_column_usage kc
7   ON tc.constraint_name = kc.constraint_name
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	table_name	constraint_name	column_name
1	baggage	fk_baggage_booking	booking_id
2	baggage_ch...	fk_baggage_check_booki...	booking_id
3	baggage_ch...	fk_baggage_check_pass...	passenger_id
4	boarding_pa...	fk_boarding_booking	booking_id
5	booking_flight	fk_booking_flight_booking	booking_id
6	booking_flight	fk_booking_flight_flight	flight_id
7	flights	fk_flights_airline	airline_id
8	flights	fk_flights_arriving_airport	arriving_airport_id
9	flights	fk_flights_departing_airp...	departing_airport...
10	security_che...	fk_security_passenger	passenger_id

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Query Query History

```
7 SELECT 'flights', COUNT(*) FROM flights
8 UNION ALL
9 SELECT 'booking', COUNT(*) FROM booking
10 UNION ALL
11 SELECT 'baggage', COUNT(*) FROM baggage
12 UNION ALL
13 SELECT 'baggage_checkl...', COUNT(*) FROM baggage_checkl...
```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	table_name	rows_count
	text	bigint
1	airline	5
2	airport	5
3	baggage	5
4	baggage_ch...	5
5	boarding_pa...	5
6	booking	5
7	booking_flight	5
8	flights	5
9	passengers	5
10	security_che...	5

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LAB WORK #2

DATABASE NORMALIZATION AND SQL IMPLEMENTATION

1. Normalization of Booking_receipt

The initial Booking_receipt table contains information about a booking, passenger, flight, airports, airline, seats and ticket price.

Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)

1.1 First Normal Form (1NF)

The main problem in the original table is seat_numbers. One cell can contain several values, for example '12A, 12B'. In 1NF every field should contain one atomic value. Therefore, the seats are separated into individual values.

Example:

Before: booking_id = 1, seat_numbers = '12A, 12B'

After: booking_id = 1, seat_number = '12A'

booking_id = 1, seat_number = '12B'

1.2 Second Normal Form (2NF)

After 1NF, the data is separated according to the objects to which it belongs. Passenger information is stored separately from booking and flight information. This removes partial dependencies.

Passenger(passenger_id PK, passenger_full_name, passenger_passport_number)

Flight(flight_id PK, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name)

Booking(booking_id PK, flight_id FK, passenger_id FK, seat_number, ticket_price)

1.3 Third Normal Form (3NF)

In 3NF, transitive dependencies are removed. For example, a booking determines a flight, and the flight determines the airline. Therefore airline information should not be stored directly in Booking. It is stored in a separate Airline table.

Airline(airline_id PK, airline_name)

Airport(airport_id PK, airport_name, city)

Flight(flight_id PK, flight_number, departure_airport_id FK, arrival_airport_id FK, airline_id FK)

Passenger(passenger_id PK, passenger_full_name, passenger_passport_number)

Booking(booking_id PK, flight_id FK, passenger_id FK, seat_number, ticket_price)

2. Functional Dependencies

- booking_id $\rightarrow$ flight_id
- booking_id $\rightarrow$ passenger_id
- booking_id $\rightarrow$ seat_number
- booking_id $\rightarrow$ ticket_price
- flight_id $\rightarrow$ airline_id
- flight_id $\rightarrow$ departure_airport_id
- flight_id $\rightarrow$ arrival_airport_id
- airline_id $\rightarrow$ airline_name
- airport_id $\rightarrow$ airport_name, city
- passenger_id $\rightarrow$ passenger_full_name, passenger_passport_number

3. Checking Flights for 3NF

The primary key of Flights is flight_id. The attributes of the flight depend on flight_id. Airline and airport information is stored in separate tables and is referenced through foreign keys.

For example: flight_id $\rightarrow$ airline_id and airline_id $\rightarrow$ airline_name. The airline_name is not stored directly in Flights, so the transitive dependency is removed from the Flights table. The same idea is used for departure and arrival airports.

Therefore, the Flights table is organized according to 3NF: non-key information about airlines and airports is kept in separate tables.

4. SQL Database Structure

The database was implemented in PostgreSQL using pgAdmin. The following tables were created:

- airline
- airport
- passengers
- flights
- booking

- baggage
- baggage_check
- boarding_pass
- booking_flight
- security_check

5. Primary Keys and Foreign Keys

Primary keys:

- airline.airline_id
- airport.airport_id
- passengers.passenger_id
- flights.flight_id
- booking.booking_id
- baggage.baggage_id
- baggage_check.baggage_check_id
- boarding_pass.boarding_pass_id
- booking_flight.booking_flight_id
- security_check.security_check_id

Main foreign-key relationships:

- security_check.passenger_id → passengers.passenger_id
- baggage_check.passenger_id → passengers.passenger_id
- baggage_check.booking_id → booking.booking_id
- baggage.booking_id → booking.booking_id
- boarding_pass.booking_id → booking.booking_id
- booking_flight.booking_id → booking.booking_id
- booking_flight.flight_id → flights.flight_id
- flights.airline_id → airline.airline_id
- flights.departing_airport_id → airport.airport_id
- flights.arriving_airport_id → airport.airport_id

6. Required Table Changes

- airline_info was renamed/represented as airline.
- booking.price was renamed to booking.ticket_price.
- flights.departing_gate was changed to TEXT.
- The unnecessary info column was not included in the final airline structure.
- All required attributes were defined as NOT NULL.

7. Sample Data

At least five sample records were inserted into each of the created tables. The data was inserted in the correct order so that foreign-key constraints were satisfied.

The sample data includes five airlines, five airports, five passengers, five flights, five bookings and related baggage, baggage checks, boarding passes, booking-flight records and security checks.

8. Screenshots

The following screenshots should be added from pgAdmin:

1. Screenshot of the list of created tables.
2. Screenshot showing the Foreign Key constraints.
3. Screenshot of the table structures/columns.
4. Screenshot of sample data in the main tables.
5. Screenshot of the SQL queries used for table creation and data insertion.

Conclusion: The database was normalized from the original Booking_receipt structure to 3NF. The final PostgreSQL database contains separate tables for airlines, airports, passengers, flights and bookings, with primary and foreign keys used to connect the data. Sample data was inserted and the relationships were checked in pgAdmin.